

Yi-Chien (Jason) Lin

✉ jasonlin316@gmail.com

in [jasonlin316](#)  [jasonlin316.github.io](#)  [jasonlin316](#)

EXPERIENCES

RESEARCH SCIENTISTS | META

07/21/2025 – Present | 1 Hacker Way, Menlo Park, CA 94025

AI Systems Co-Design team member focused on performance projection for GenAI training on large-scale GPU clusters. I help evaluate next-generation LLM architectures through performance modeling, including estimating achievable MFU on target clusters, to identify system-efficient design candidates.

RESEARCH INTERN | MICROSOFT

05/20/2024 – 08/16/2024 | 1045 La Avenida St, Mountain View, CA 94043

Worked with the AI Framework Team on developing a solution for automatic model parallelization for LLM serving systems. GitHub available: [microsoft/apex_plus](#).

EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA | PH.D. IN ELECTRICAL ENGINEERING

August 2020 - June 2025 | Los Angeles, CA • Dissertation: Automated Graph Machine Learning Systems

UNIVERSITY OF SOUTHERN CALIFORNIA | M.S. IN ELECTRICAL ENGINEERING

August 2020 - May 2024 | Los Angeles, CA • Cum. GPA: 3.88/4.0

NATIONAL TAIWAN UNIVERSITY | B.S. IN ELECTRICAL ENGINEERING

September 2016 - June 2020 | Taipei, Taiwan • Cum. GPA: 3.75/4.3

PUBLICATIONS

- [1] Yi-Chien Lin, Yuyang Chen, Sameh Gobriel, Nilesh Jain, and Viktor Prasanna. *ARGO+: Achieving Multi-Level Scalable GNN Training on Distributed Multi-Core Platform*. IEEE Transactions on Parallel and Distributed Systems (TPDS), 2026.
- [2] Yi-Chien Lin, Haoyang Fan, Sameh Gobriel, Nilesh Jain, and Viktor K. Prasanna. *Accelerating GNN Inference via Automated Parallel Execution on Edge Heterogeneous Platforms*. In International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD), 2025.
- [3] Rakshith Jayanth, Yi-Chien Lin, Souvik Kundu, Deepak A. Mathaikutty, and Viktor Prasanna. *ACCLAIM: Accelerating Long Context LLM Inference on Heterogeneous Edge Platforms*. In The Latin America High Performance Computing (CARLA), 2025.
- [4] Haoyang Fan, Yi-Chien Lin, and Viktor Prasanna. *ELLIE: Energy-Efficient LLM Inference at the Edge via Prefill-Decode Splitting*. In International Conference on Application-specific Systems, Architectures and Processors (ASAP), 2025.
- [5] Sachini Wickramasinghe, Yi-Chien Lin, Cauligi Raghavendra, and Viktor Prasanna. *SMART: High-Performance SAR ATR Through Model-Architecture Co-Design on FPGA*. In International Symposium on Field-Programmable Custom Computing Machines (FCCM), 2025.
- [6] Yi-Chien Lin, Woosuk Kwon, Ronald Pineda, and Fanny Nina Paravecino. *APEX: An Extensible and Dynamism-Aware Simulator for Automated Parallel Execution in LLM Serving*. In arXiv preprint arXiv:2411.17651, 2024.
- [7] Yi-Chien Lin, Zhijie Xu, and Viktor Prasanna. *xBS-GNN: Accelerating Billion-Scale GNN Training on FPGA*. In Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis (SC-W), 2024.
- [8] Yi-Chien Lin, Gangde Deng, and Viktor Prasanna. *A Unified CPU-GPU Protocol for GNN Training*. In ACM International Conference on Computing Frontiers (CF), 2024.
- [9] Yi-Chien Lin, Yuyang Chen, Sameh Gobriel, Nilesh Jain, Gopi Krishna Jhaand, and Viktor Prasanna. *ARGO: An Auto-Tuning Runtime System for Scalable GNN Training on Multi-Core Processor*. In IEEE International Parallel and Distributed Processing Symposium (IPDPS), 2024 [Best Paper Nominee].

- [10] Yi-Chien Lin, Bingyi Zhang, and Viktor Prasanna. *HitGNN: High-throughput GNN Training Framework on CPU+Multi-FPGA Heterogeneous Platform*. In IEEE Transactions on Parallel and Distributed Systems (TPDS), 2024.
- [11] Yi-Chien Lin and Viktor Prasanna. *HyScale-GNN: A Scalable Hybrid GNN Training System on Single-Node Heterogeneous Architecture*. In IEEE International Parallel & Distributed Processing Symposium (IPDPS), 2023.
- [12] Yi-Chien Lin, Bingyi Zhang, and Viktor Prasanna. *Accelerating GNN Training on CPU+Multi-FPGA Heterogeneous Platform*. In Latin American High Performance Computing Conference (CARLA), 2022.
- [13] Yi-Chien Lin, Bingyi Zhang, and Viktor Prasanna. *HP-GNN: Generating High Throughput GNN Training Implementation on CPU-FPGA Heterogeneous Platform*. In ACM International Symposium on Field-Programmable Gate Arrays (FPGA), 2022.
- [14] Yi-Chien Lin, Bingyi Zhang, and Viktor Prasanna. *GCN Inference Acceleration using High-Level Synthesis*. In IEEE High Performance Extreme Computing Conference (HPEC), 2021.
- [15] Jing-Ping Wu, Yi-Chien Lin, Ying-Wei Wu, Shih-Wei Hsieh, Ching-Hsuan Tai, and Yi-Chang Lu. *A Memory-Efficient Accelerator for DNA Sequence Alignment with Two-Piece Affine Gap Tracebacks*. In IEEE International Symposium on Circuits and Systems (ISCAS), 2021.

PATENTS

- [1] Yi-Chien Lin and Chih-Hung Lin and Ling-Yu Wu and Chih-Shiuan Lee. *Face Orientation-Based Cursor Positioning on Display Screens*. In WIPO Patent No. WO/2021/145855), 2021.

SERVICES

WORKSHOP CHAIR | THE 31ST IEEE HIPC CONFERENCE
2024 | Bengaluru, India

REGISTRATION CHAIR | THE 31ST IEEE FCCM CONFERENCE
May 2023 | Los Angeles, CA

CONFERENCE/JOURNAL REVIEWER IEEE IPDPS, ACM FPGA, SC, IEEE ASAP
2020 - Present

TEACHING ASSISTANT

- EE451: Parallel and Distributed Computation (Spring '21, Fall '22)
- EE457: Computer Systems Organization (Fall '21)

AWARDS

CAPITAL ONE FELLOW
USC-Capital One Center for Responsible AI and Decision Making in Finance (CREDIF). 2024

SRC AIHW ANNUAL REVIEW BEST PRESENTATION AWARD
Wilmington, MA. 2024

IPDPS BEST POSTER AWARD HONORABLE MENTION
San Francisco, CA. 2024

VITERBI SCHOOL OF ENGINEERING FELLOWSHIP
Los Angeles, CA. 2020

NTUEE UNDERGRADUATE INNOVATION AWARD THIRD PRIZE
Taipei, Taiwan. 2019

SKILLS

ML SYSTEMS HARDWARE/SOFTWARE CO-DESIGN • PERFORMANCE MODELING • PYTORCH • PTHREADS • MPI • OPENMP • CUDA

HARDWARE PROGRAMMING VERILOG/SYSTEMVERILOG • HIGH-LEVEL SYNTHESIS (HLS)